Lab Session

Task 1: Find the optimal solution of each of the following problems.

Problem 1:

$$
\begin{aligned}
\text{Minimize} \quad & x_1 + x_2 \\
\text{subject to} \quad & x_1 + x_2 \geq 1 \\
& \mathbf{0} \leq \mathbf{x} \leq \mathbf{1}
\end{aligned}
$$

Problem 2:

$$
\begin{aligned}
\text{Minimize} \quad & x_1 + x_2 + x_3 + x_4 \\
\text{subject to} \quad & x_1 + x_2 = 1 \\
& x_2 + x_3 = 1 \\
& x_3 + x_4 = 1 \\
& x_1 + x_4 = 1 \\
& \mathbf{0} \leq \mathbf{x} \leq \mathbf{1}
\end{aligned}
$$

Task 2: Choose three LP solvers with different algorithms and/or different languages. Then solve the above two problems using each solver.

Lab Session

Task 3: Compare the following four methods on your 100-item knapsack problem (used in the lab session on Simulated Annealing) with respect to the solution quality (i.e., the objective function value), and the total computation time.

(i) A heuristic method used to generate an initial solution for SA.

(ii) Your simulated annealing algorithm with your parameter setting.

(iii) Use of an LP solver (i.e., use of the LP relaxation problem), and create a feasible solution from the LP solution.

(iv) Use of an integer LP solver (ILP solver).

Task 4: You will receive 200-item and 400-item knapsack problems from our TA. Compare the above-mentioned four algorithms on each of those two problems in the same manner as in Task 3.

Goal: To understand the difference between LP and ILP solvers, and to understand the dependency of the computation time of each algorithm on the problem size.